

CLAYTON, NM, USA

WMO: 723600

Lat: 36.449N

Lon: 103.154W

Elev: 4960

StdP: 12.25

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23051

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	8.3	13.7	-4.6	5.2	31.7	-0.6	6.4	35.2	35.2	33.9	31.8	37.6	10.0	30	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.5	95.8	62.6	93.4	62.6	90.9	62.6	68.2	84.1	66.9	83.3	65.9	82.4	15.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	107.5	71.0	62.6	102.3	70.2	61.3	97.7	69.9	35.3	83.9	34.2	83.0	33.3	82.0	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
30.8	26.9	24.3	0.8	99.8	4.7	2.1	-2.6	101.3	-5.3	102.5	-7.9	103.7	-11.3	105.3		
			-0.1	70.9	4.6	4.3	-3.4	74.0	-6.1	76.6	-8.6	79.0	-11.9	82.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.2	36.4	38.4	45.7	52.7	62.2	72.0	76.0	74.1	67.3	55.6	45.1	36.1
	DBStd	16.73	10.22	10.61	10.14	9.24	8.22	6.21	4.70	4.92	7.05	9.07	9.89	10.25
	HDD50	1754	428	342	201	79	10	0	0	0	2	49	204	439
	HDD65	4632	885	746	600	375	151	14	2	3	57	307	596	896
	CDD50	3662	8	16	67	159	387	661	807	747	521	223	58	8
	CDD65	1065	0	0	1	6	63	225	344	285	126	15	0	0
	CDH74	12930	1	9	59	232	1070	2845	3898	3001	1460	325	30	0
	CDH80	5931	0	0	5	36	394	1419	2020	1421	566	69	1	0
Wind	WSAvg	12.2	11.7	12.4	13.0	14.0	13.0	12.7	11.4	10.6	11.6	12.0	12.1	11.8
Precipitation	PrecAvg	15.4	0.3	0.3	0.6	0.9	2.0	2.3	2.6	2.7	1.9	0.9	0.6	0.3
	PrecMax	25.7	1.1	1.6	2.3	3.1	4.4	5.0	6.7	5.8	5.2	4.6	2.2	1.1
	PrecMin	8.9	0.0	0.0	0.0	0.0	0.3	0.5	0.2	0.3	0.4	0.0	0.0	0.0
	PrecStd	4.0	0.3	0.4	0.5	0.8	1.1	1.1	1.5	1.5	1.3	1.3	0.6	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.6	73.9	79.7	84.3	92.7	99.0	98.6	97.0	93.7	86.2	77.8	69.4
		MCWB	45.5	47.6	50.1	52.4	59.1	60.5	63.0	64.2	60.5	55.8	49.9	45.5
	2%	DB	64.3	67.4	75.0	79.9	88.3	95.5	95.7	94.0	89.8	81.6	73.2	63.1
		MCWB	43.3	44.9	48.3	51.2	56.8	60.7	63.3	63.8	60.2	54.9	48.1	43.2
	5%	DB	58.9	62.9	70.2	76.4	84.6	92.3	93.3	91.3	86.6	77.8	68.0	58.2
		MCWB	41.3	42.7	46.4	50.1	55.9	60.6	63.8	63.9	60.2	53.8	46.6	40.9
	10%	DB	53.5	57.2	65.0	72.1	80.4	88.8	90.7	88.3	82.9	72.7	62.4	53.1
		MCWB	38.4	40.5	44.7	49.2	55.1	60.9	64.0	64.0	60.2	52.0	44.5	38.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.8	48.3	52.7	57.7	65.2	68.3	71.1	70.2	67.0	62.6	53.1	46.9
		MCDB	66.8	70.3	70.5	72.7	84.0	86.5	87.0	85.1	80.2	71.4	66.5	64.8
	2%	WB	44.1	45.7	49.9	54.9	61.5	66.1	68.2	68.1	64.8	59.3	50.2	44.0
		MCDB	62.2	64.6	70.1	70.3	76.7	83.7	83.9	83.9	78.9	70.5	67.9	61.7
	5%	WB	41.6	43.4	47.7	52.5	59.5	64.6	67.0	66.8	63.5	56.7	47.9	41.5
		MCDB	58.0	61.6	67.4	69.5	75.3	81.8	83.5	82.8	77.5	69.8	65.1	56.5
	10%	WB	39.1	41.0	45.7	50.3	57.6	63.4	66.0	65.7	62.1	54.3	45.6	39.0
		MCDB	52.4	56.3	63.7	68.3	73.3	80.3	83.1	81.6	76.4	69.1	61.6	52.1
Mean Daily Temperature Range	5% DB	MDBR	25.1	26.2	27.4	27.3	27.1	27.7	26.5	24.8	25.3	25.8	25.9	23.7
		MCDBR	32.3	34.1	33.6	33.9	32.4	32.8	30.0	29.3	30.2	31.6	32.6	30.7
		MCWBR	17.1	17.1	15.2	15.0	13.3	11.4	8.8	9.0	10.2	13.0	15.4	16.1
	5% WB	MCDBR	30.9	32.3	31.6	29.4	28.7	27.9	25.8	25.3	24.9	25.9	29.7	29.7
		MCWBR	17.0	16.7	15.2	14.5	13.6	11.1	8.9	9.6	9.5	12.5	14.9	16.2
Clear-Sky Solar Irradiance	taub	0.241	0.247	0.276	0.298	0.317	0.342	0.364	0.347	0.313	0.280	0.255	0.242	
	taud	2.603	2.610	2.515	2.458	2.465	2.425	2.419	2.461	2.524	2.587	2.574	2.594	
	Ebn at Noon	312	321	317	312	306	296	289	293	300	304	303	305	
	Edh at Noon	23	26	31	35	35	36	36	34	30	26	23	22	
All-Sky Solar Radiation	RadAvg	968	1255	1658	2037	2318	2453	2351	2109	1811	1389	1052	834	
	RadStd	73	88	127	109	133	108	87	95	92	112	73	47	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (17 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	N/A	

Nomenclature: See separate page